Trainings

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : Determine the appropriate radiator or coolant expansion tank cap adapter for the vehicle being serviced. A list of available caps and applications can be found in Section 14 of the Service Products Catalog, Bulletin 3377710.

Use a vacuum pump type device on the cooling system. Install the adapter on the radiator or coolant expansion tank. Connect an automotive hand held vacuum pump and apply 3 in Hg [40 in H_2O] of vacuum.

Note : Depending on the cooling system configuration, more vacuum may be required to keep the cooling system in balance.

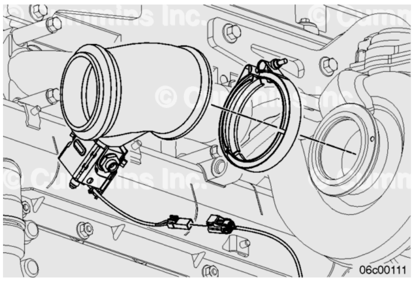
Place a container under the aftertreatment fuel injector. There may be a small quantity of coolant that drains from the coolant lines when disconnected.

- If a vacuum type device is **not** available to properly hold coolant in the engine during servicing, the coolant may be drained.
    - For Signature™, ISX, and QSX15 engines, use the following procedure in the Signature™, ISX, and QSX15 Service Manual, Bulletin 3666239. Refer to Procedure 008-018 in Section 8.

(/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
    - For ISM, ISMe, and QSM11 engines, use the following procedure in the ISM, ISMe, and QSM11 Service Manual, Bulletin 3666322. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Disconnect the fuel lines and coolant lines **only** if removing the injector from the engine for cleaning or replacement. Refer to Procedure 011-051 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html)
- Disconnect the wiring harness from the aftertreatment injector.
- Remove the adapter tube and injector as an assembly. Refer to Procedure 011-043 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html)

Remove

Remove the adaptor tube gasket.



Remove the capscrews holding the injector.

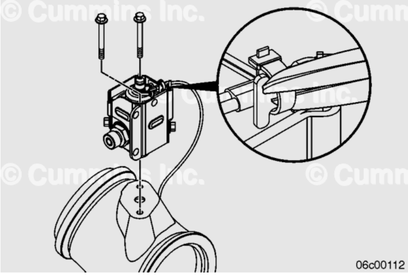
Remove the injector assembly.

Cut the EZ tie holding the injector harness to the heat shield.

Discard the metal gasket.

Discard the fibrous insulator.

Discard the capscrews



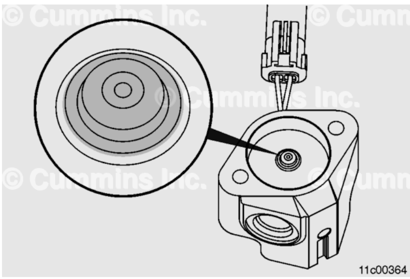
Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

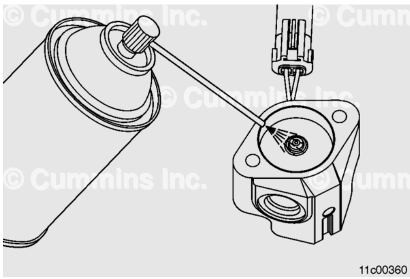
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Inspect the injector tip.

Some carboning can occur on the tip of the injector.

Spray the tip of the aftertreatment fuel injector with carburetor cleaner. When spraying, focus the spray on the tip of the injector. Use enough carburetor cleaner to fill the cavity surrounding the tip and to cover the tip.

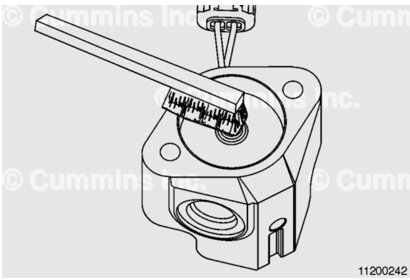


Allow the carburetor cleaner to sit for 15 seconds so that it can break down and penetrate the debris that has collected in the tip area.

Note : Carburetor cleaner is the most effective solvent for this procedure. Do not substitute any other cleaning solvent.

⚠ CAUTION ⚠

Use only a stiff brass brush with undamaged bristles to clean the tip of the aftertreatment fuel injector.



⚠ CAUTION ⚠

The use of a brush with damaged bristles will not be as effective.

⚠ CAUTION ⚠

The use of a steel wire brush or a steel wire wheel will cause permanent damage to the aftertreatment fuel injector.

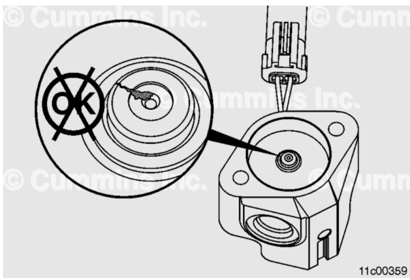
Use a brass brush to agitate the carburetor cleaner, focusing on the tip of the injector.

Repeat spraying and brushing for 1 minute. Often the carbon will be removed from the tip in a short period, but cleaning for a full minute will remove the carbon built up in the nozzle.

Note : The carbon this procedure is attempting to remove can often not be seen with the unaided eye. This carbon builds up in the nozzle tip area of the injector.

Inspect the injector tip for cracks or other damage. Replace if necessary.

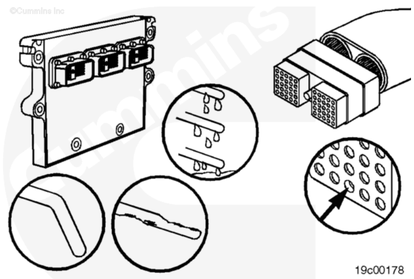
Clean the adapter tube injector mounting surface. Refer to Procedure 011-043 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html)



Inspect the harness and connector pins for the following:

- Loose connector
- Corroded pins
- Bent or broken pins
- Pushed back or expanded pins
- Moisture in or on the connector
- Missing or damaged connector seals
- Dirt or debris in or on the connector pins
- Connector shell broken
- Wire insulation damage
- Damaged connector locking tab.

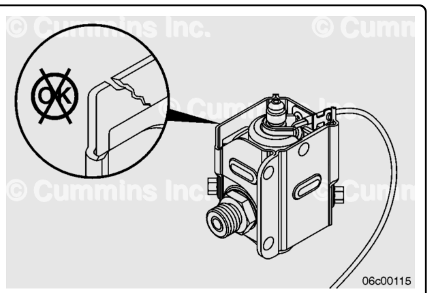
Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)



Inspect the heat shield, if applicable, for the following:

- Cracks
- Dents.

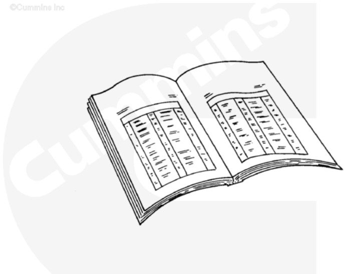
Replace if necessary.



Inspect for Reuse

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



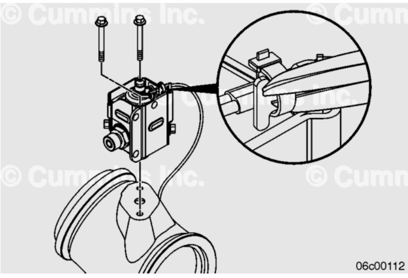
- Connect the fuel line and coolant lines. Refer to Procedure 011-051 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html)
- Connect the adapter pipe to the turbocharger. Refer to Procedure 011-043 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html)
- Connect the wiring harness to the aftertreatment injector.
- If the coolant was drained, fill the cooling system.
    - Use the following procedure in the Signature™, ISX, and QSX15 Service Manual, Bulletin 3666239. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
    - Use the following procedure in the ISM, ISMe, and QSM11 Service Manual, Bulletin 3666322. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- If a vacuum pump was used, remove the vacuum pump.
- Start the engine and check for leaks.

Note : If a malfunction resulted in coolant entering the exhaust system, the aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-013.html)

Note : If a malfunction resulted in resulted in oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Refer to the Aftetreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter

Reuse Guidelines, Bulletin 4021600. (/qs3/pubsys2/xml/en/bulletin/4021600.html)

Install



Install a new fibrous insulator.

Install a new metal gasket.

Note : The fibrous insulator will be against the injector.

Install the aftertreatment injector onto the adapter pipe.

Note : Make sure the aftertreatment injector harness is not touching the adapter pipe.

Note : The tall side of the injector and injector heat shield should be facing the turbocharger side.

Install new capscrews. Anti-seize compound will already be applied to the capscrews.

The use of the original capscrews will result in clamp load loss and gasket leakage.

Torque Value: 11.3 n•m [100 in-lb]

Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Connect the fuel line and coolant lines. Refer to Procedure 011-051 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html)
- Connect the adapter pipe to the turbocharger. Refer to Procedure 011-043 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html)
- Connect the wiring harness to the aftertreatment injector.

- If the coolant was drained, fill the cooling system.
    - Use the following procedure in the Signature™, ISX, and QSX15 Service Manual, Bulletin 3666239. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
    - Use the following procedure in the ISM, ISMe, and QSM11 Service Manual, Bulletin 3666322. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- If a vacuum pump was used, remove the vacuum pump.
- Start the engine and check for leaks.

Note : If a malfunction resulted in coolant entering the exhaust system, the aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)

Note : If a malfunction resulted in resulted in oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Refer to the Aftetreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600. (/qs3/pubsys2/xml/en/bulletin/4021600.html)

Last Modified: 21-Mar-2017